# NAYAN

**COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE DEVELOPMENT | AI/ML | CYBERSECURITY**

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## PROFESSIONAL PROFILE

Computer Science & Engineering (Cyber Security) undergraduate (B.E., expected 2028), published patent-application co-inventor, and hands-on builder of software, AI/ML, privacy, and cybersecurity solutions. Experience spans research, AI internships, shipped web products, and student-led technical programs. Indian Army Terrier Cyber Quest 2026 Top-10 national finalist and innovation award winner with cross-functional communication, product ownership, and rapid prototyping.

## EDUCATION

**B.E., Computer Science & Engineering (Cyber Security) | Aarupadai Veedu Institute of Technology, Chennai** 2024-2028

CBSE Class XII: St. John's Academy, Mahua, Bihar | CBSE Class X: B.D. Public School, Hajipur (Panapur Langa), Bihar

## CORE TECHNICAL SKILLS

**Programming & Web:** Python, PHP, JavaScript, SQL, HTML, CSS

**AI, Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, VGG16, Faster R-CNN, Federated Learning, retrieval-augmented generation (RAG), LangChain, Llama-3, GPT-4o

**Application Development:** Flask, FastAPI, React, Tailwind CSS, OpenCV, Tkinter, Streamlit, Jupyter, VS Code

**Cybersecurity & Privacy: Digital forensics, phishing analysis, Kusto Query Language (KQL), OSINT, email/log investigation, TOR traffic analysis, IoT security, ASCON cryptography, data redaction and audit trails**

**Data & Retrieval:** MySQL, MongoDB, Neo4j, Pinecone

**Additional:** Google Cloud Platform, Intel oneAPI/OpenVINO, TinyML, GenAI, agentic AI, ASCON, Cardano/Plutus, zk-SNARKs, technical presentations and teamwork

## EXPERIENCE AND LEADERSHIP

**Summer Research Intern (Selected) | TATA Centre for AI & ML, NIT Tiruchirappalli** May-Jul 2026

- Selected for a six-week Education Track research program following first place at the AI For ALL Ideathon; led Team PARTH on SUTRADHAR under faculty mentorship.

**AI/ML Intern | iTech (AVIT Hackathon 2026 Award)** 2026-Present

- Awarded an AI/ML internship after winning AVIT Hackathon 2026, held 15 July 2026 by AVIT in partnership with C3 as an iTech initiative.

**Agentic AI Intern | Brain O Vision Solutions India Pvt. Ltd.** Jan 2026-Present

- Developing agentic-AI workflows for process automation and decision support in organizational settings.

**R&D Intern, Campus Catalyst Program (CCIP 2026) | Brain O Vision Solutions India Pvt. Ltd.** Feb 2026-Present

- Contributing to applied AI research and automation in the R&D business division (Intern ID: BOV-CCIP-002).

**Web Developer | Eduooze, School of Banking** 2025-Present

- Developed and maintained a customer-facing banking EdTech portal covering enrollment, placement, and product sales using HTML, CSS, JavaScript, PHP, and MySQL.

**Vice President | CYFENCE Cybersecurity Club, AVIT** 2026-Present

- Lead technical events, security awareness sessions, workshops, and faculty-student collaboration for the cybersecurity community.

Additional leadership: Student Ambassador, AVIT (2025-2026); Google Ambassador (2025-Present); Innovation Ambassador, MIC & AICTE (2025); Team Lead, PARTH.

## PUBLISHED PATENT APPLICATION

Co-Inventor, Adaptive AI-Lightweight Crypto Framework (ALCF) for Real-Time IoT Security | Indian Patent Application 202541117496 A; filed Nov 2025, published in The Patent Office Journal 51/2025 (Dec 2025). Device/Edge/Cloud system using ASCON and TinyML; documented prototype targets include <2 s detection and 30-50% energy savings.

## SELECTED PROJECTS

**Result Analyzer | An FA's Pocketeer** | ***Full-Stack Product / Education Analytics***

- Independently developed an academic-results application with Admin/Faculty/Student portals, bulk Excel uploads, automated SGPA/CGPA calculations, grading analytics, and university-aligned workflows.
- Champion, AVIT Mini Product Showcase 2026; Certificate of Appreciation from AVIT CSE Department for implementing the Result Analysis Generator App. Live: resultanalyzer.infinityfreeapp.com.

**JobDham.in | Full-Stack Job Portal** | ***Web Development / Live Product***

- Built and hosted a job platform from scratch using HTML, CSS, JavaScript, PHP, MySQL, and Hostinger.
- Integrated Google verification and AdSense monetization; operated as an advertising-supported, revenue-generating product. Live: jobdham.in.

**Project SUTRADHAR | Context-Aware Generative AI Tutor for Bharat** | ***GenAI / EdTech / NIT Tiruchirappalli***

- Led Team PARTH; built the Gurukul Context Engine to adapt NCERT examples into local analogies using Llama-3 and the Bhashini API.
- Designed regional-dialect, lip-synced tutoring with Stable Diffusion XL/LoRA, Wav2Lip, and Tortoise TTS; supported low-bandwidth WhatsApp/SMS learning on 2G/3G. Related recognition: AI For ALL Ideathon first place.

**TOR Sentinel | Privacy-Preserving Federated AI for TOR Traffic Analysis** | ***Cybersecurity / Federated Learning***

- Designed federated analysis across ISPs without sharing raw personally identifiable data, using Adaptive Time-Window Correlation (ATWC) to address variable TOR latency.
- Reported >80% probabilistic tracing confidence in simulation, with blockchain-based judicial evidence trails and simulated investigation times reduced from weeks to hours. Associated with Innoventia 2K26 and the TN Police Cyber Crime Hackathon.

**ID Masker Tool | AI-Powered PII Detection & Redaction** | ***Privacy Engineering / Computer Vision***

- Trained a Faster R-CNN (ResNet-50) image model to detect sensitive information; implemented black-box, Gaussian blur, and pixelation redaction.

**AEGIS | Adaptive Enterprise Guardrail & Intelligence System** | ***GenAI Security / ET GenAI Hackathon***

- Designed GhostGuard, a shift-left semantic firewall against package-name slopsquatting using an NPM/PyPI knowledge graph, and Chimera, a runtime adaptive-deception/digital-twin engine.

**BharatRealEstateAI | AI Real Estate Discovery** | ***Applied AI / Intel Unnati Program***

- Designed emotion-aware RAG property discovery with semantic search in 12+ Indian languages, Vastu and joint-family cultural preferences, and visual similarity matching.

**NutriLens | AI-First Food & Health Coaching** | ***HealthTech / AMD Slingshot Hackathon***

- Architected an AI nutrition/coaching application incorporating Indian regional cuisine and 8+ Google Cloud services.

**EchoShield | AI Voice Deepfake & Vishing Defense** | ***Audio Security / Edge AI***

- Developed a privacy-first, on-device voice-liveness detection approach using breathing, micro-jitters, and room-tone signals; reported 98.7% accuracy and <1.2 s response time.

**Sanjivani | Decentralized Offline-First Health Ecosystem** | ***HealthTech / Pysellers Innovation Challenge***

- Designed an offline-first health ecosystem with quantized Phi-3/TinyLlama edge AI, multi-agent triage, patient-owned DID/AES-256 records, federated learning, Wi-Fi Aware mesh networking, and a Health Points economy.

**SOVARTH / Sovereign Nexus | Privacy-First Cardano DeFi for Asia** | ***Blockchain / Cardano Hackathon Asia 2025***

- Led Team PARTH in designing an identity, asset, credit, and cross-chain wallet concept for underserved Asian users; ranked among the top 112 teams in Asia.

## SELECTED RECOGNITION & CREDENTIALS

- Top-10 National Finalist — Indian Army Terrier Cyber Quest 2026 (AI Kavach, Track 2). Grand Finale, New Delhi; selected from a national field of 8,200+ teams. Certificate of Appreciation for exemplary performance from the Director General, Territorial Army (Oct 2026).

Young Tech Visionary Award (Innoventia 2K26) | 1st Place, AI For ALL Ideathon (NIT Tiruchirappalli) | 2nd Place & Top 10 Finalist, Tamil Nadu Police Cyber Crime Hackathon 2025 | Top 112 Teams (Asia), Cardano Hackathon Asia 2025 | Winning Team, AVIT Hackathon 2026 (iTech/C3; AI/ML internship awarded) | Champion, AVIT Mini Product Showcase 2026.

Innovation Ambassador (MIC & AICTE, Advanced/Foundation) | Google Developer Design-to-Code | MongoDB SQL-to-Document Model | Full Stack Software Development (20 hrs) | DSA for DeepTech Bootcamp (2026)

Live work: jobdham.in | resultanalyzer.infinityfreeapp.com | eduooze.in

Hands-on cybersecurity labs: KC7 Cyber Detective Game - Clout Defender and Valdorian Times badges (2026); investigated phishing, threat activity and logs using KQL.